

FORCE-AWARE PEG INSERTION PROJECT

TensorBoard Guide

Every Graph Explained

Baseline (force-aware) vs. Policy A (geometry-only) ablation
20-iteration smoke-test runs · 64 parallel environments
DGX Spark (aarch64, NVIDIA GB10) · 2026-08-08

Run	Task ID	Log directory
Baseline	Shaurya-ForcePegInsert-Direct-v0	logs/rl_games/ Forge/2026-08-08_01-20-14/
Policy A	Shaurya-ForcePegInsert-PolicyA-Direct-v0	logs/rl_games/ Forge/2026-08-08_01-34-29/

Overview

Both runs were 20-iteration smoke tests — enough to prove the pipeline runs and the ablation actually removed what it was supposed to, **not** enough for the policy to have learned to insert the peg. Treat every curve here as “does this look like a healthy training process just getting started,” not “is this a good policy.”

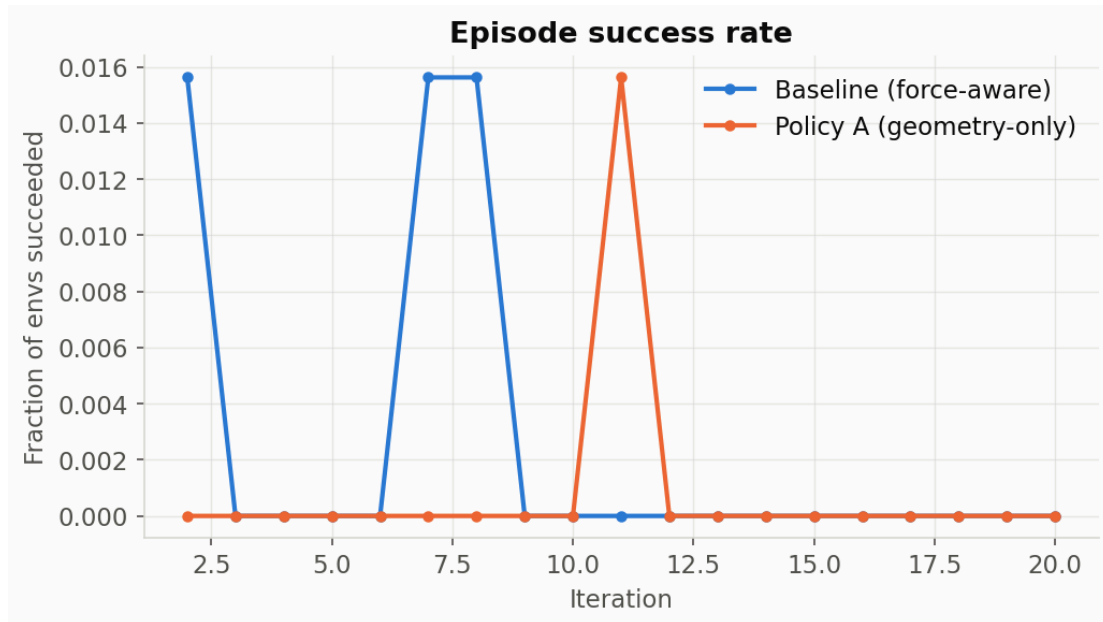
TensorBoard itself runs locally on the training machine at <http://localhost:6006/>, watching `force_peg_rl/logs/rl_games` with a 5-second reload interval — open that directly in a browser to see these same curves live and interact with them (zoom, smoothing slider, toggle runs on/off). The charts in this report are static renders of the exact same underlying data, generated with matplotlib directly from the `events.out.tfevents.*` files so they are guaranteed to match what TensorBoard shows.

Every scalar in these runs is logged three times, as `<name>/iter`, `<name>/step`, and `<name>/time` — same values, three different x-axis units (training iteration, environment step count, and wall-clock seconds). This report always uses `/iter` since that is the most useful axis for comparing two runs of the same length; the underlying number is identical on all three.

Color key used throughout this report: Baseline — force-aware draws every curve in blue; Policy A — geometry-only draws every curve in orange. Single-run charts (marked in their section) show baseline data only.

1. Episode success rate

Tag: Episode/Metrics/success_rate



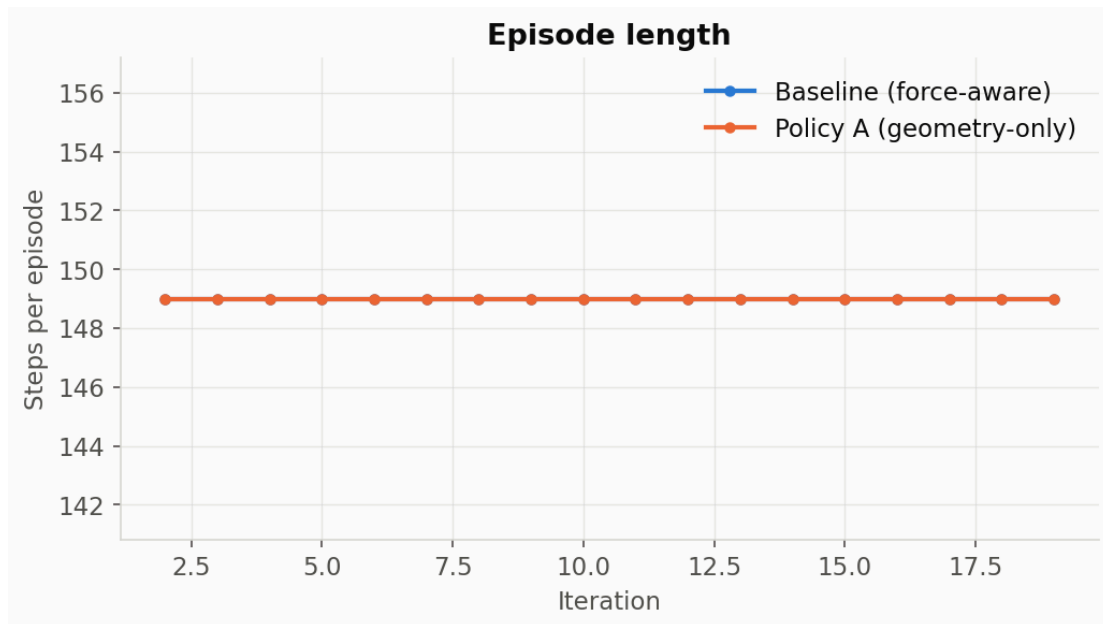
What it is: the fraction of the 64 parallel environments that met the task's success condition (`_get_curr_successes()` in `factory_env.py` — peg centered within 0.0025 xy-tolerance and below a height threshold) at the moment their episode ended, averaged per training iteration. Comes from `FactoryEnv._log_factory_metrics()`:

```
self.extras.setdefault("log", {})["Metrics/success_rate"] = (
    torch.count_nonzero(curr_successes) / self.num_envs
).item()
```

What we see: both runs sit at or near 0.0 for almost every iteration — at most 1/64 envs succeeding in any given iteration. That's expected: this is a 20-iteration smoke test, and PPO needs thousands of iterations to learn contact-rich insertion from scratch. This graph is the one to watch climb toward 1.0 over a real multi-thousand-iteration run — if it's still flat at zero after a few hundred iterations in a real run, that's a sign something is wrong (reward not shaped correctly, task too hard for the current curriculum stage, etc).

2. Episode length

Tag: episode_lengths/iter



What it is: average number of physics steps each episode ran before reset.

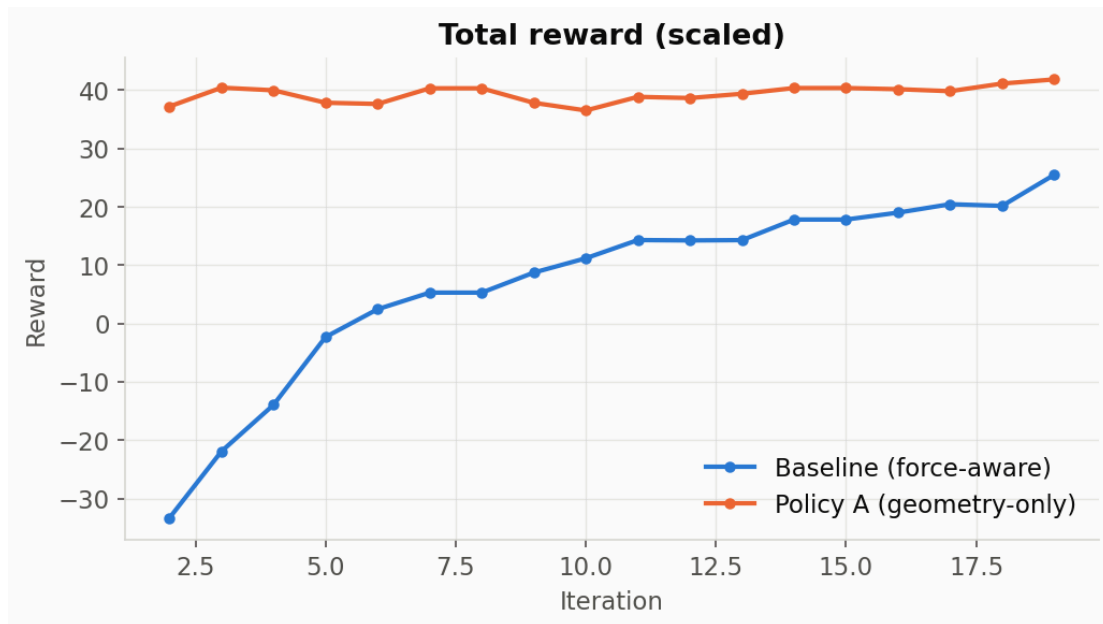
What we see: flat at exactly 149 for every iteration, in both runs. This is expected and, in this codebase, load-bearing: `_get_dones()` (our fork's override, see `docs/experiment_log.md`) keeps every reset synchronized — all 64 envs finish and reset together, always at the same fixed timeout (`episode_length_s=10.0` combined with the physics/decimation settings yields 149 steps). Episodes always run to the full timeout — early success and force-limit conditions are **labeled** but don't end the episode. If you ever see this line move, something upstream of our fork changed the timeout config or the sync-reset assumption was relaxed.

Early termination (which project §9.6 does call for) is **not yet implemented, and not proven impossible** — an initial attempt crashed on a one-line shape bug in upstream's `randomize_initial_state()` (`factory_env.py:659, a[:]` full-slice assignment), which was mistakenly read at the time as evidence that per-env reset was infeasible. See the correction block in `docs/experiment_log.md` for what the traceback actually shows. Treat the flat 149 here as a current property of this fork, not a settled architectural limit.

A field this graph implies but doesn't show — `termination_reason`: Our `ForgeEnv._get_dones()` computes a per-env label ("success", "timeout", "force_limit", or "other") every step and writes it to `self.extras["termination_reason"]` as a numpy string array. It's **not** a TensorBoard scalar (strings aren't plottable as a time series) — it's there for a future evaluation script to read from logged trajectories, per `force_aware_peg_insertion_project.md` §9.6's logging requirement. Right now, with episode length pinned at 149 for every env, the label will read "timeout" for nearly all resets and occasionally "success"/"force_limit" for envs that crossed those conditions before the timeout — worth checking once real training starts producing more successes.

3. Total reward (scaled)

Tag: rewards/iter

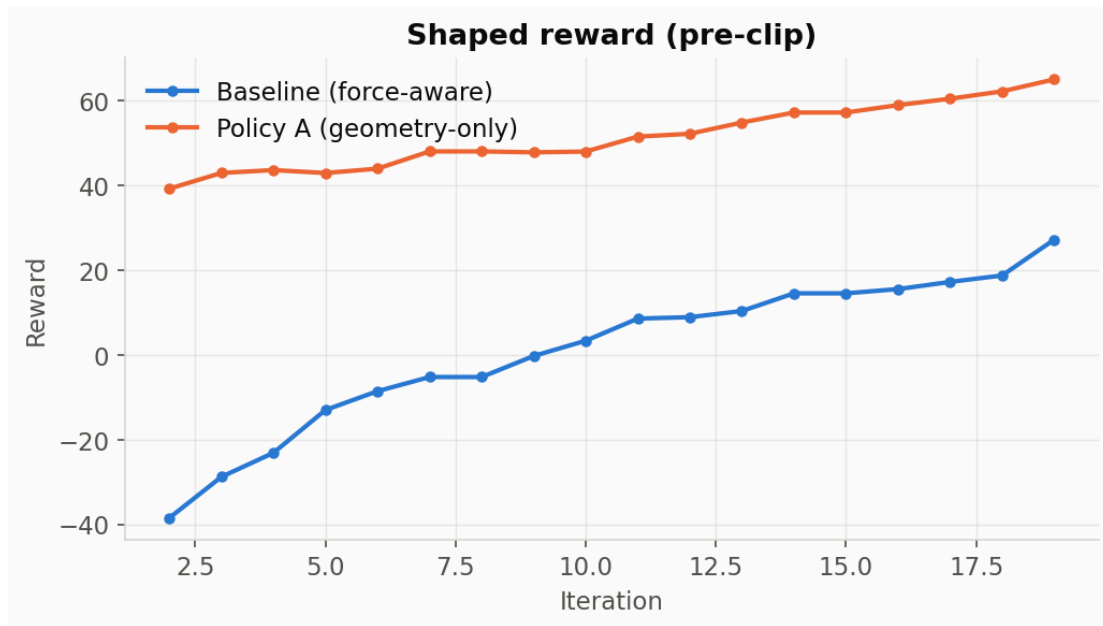


What it is: `rew_buf` — the actual sum RL-Games optimizes against, after all reward-scale multipliers (`rew_scales` dict) have been applied and RL-Games’ own observation/reward normalization wrapper has run. This is the single number that matters most for “is the policy improving,” but per the project doc’s own rule (§9.4: “Log every reward component separately. Never debug only the total reward”), it should never be read in isolation from the component breakdowns further in this report.

What we see: the baseline climbs steadily from about -33 to $+25$ over 20 iterations — classic early-training behavior, moving in the right direction. Policy A starts already around $+37$ and stays roughly flat around $+38$ to $+42$. This gap is expected and not evidence Policy A is “better”: Policy A has zero contact penalty (see §5) and a smaller, easier observation space, so its reward floor sits higher from iteration 1. Comparing the **absolute** reward level between an ablation and the baseline is not meaningful — what matters is each curve’s own trend and the component breakdown, not a head-to-head number.

4. Shaped reward (pre-clip)

Tag: `shaped_rewards/iter`



What it is: RL-Games' own internal reward-shaping/normalization output — this is what the **value function** is trained against, distinct from `rewards/iter` which is closer to the raw environment reward. The two tend to move together but are not identical (this comes from RL-Games' `a2c_common.py`, not our env code).

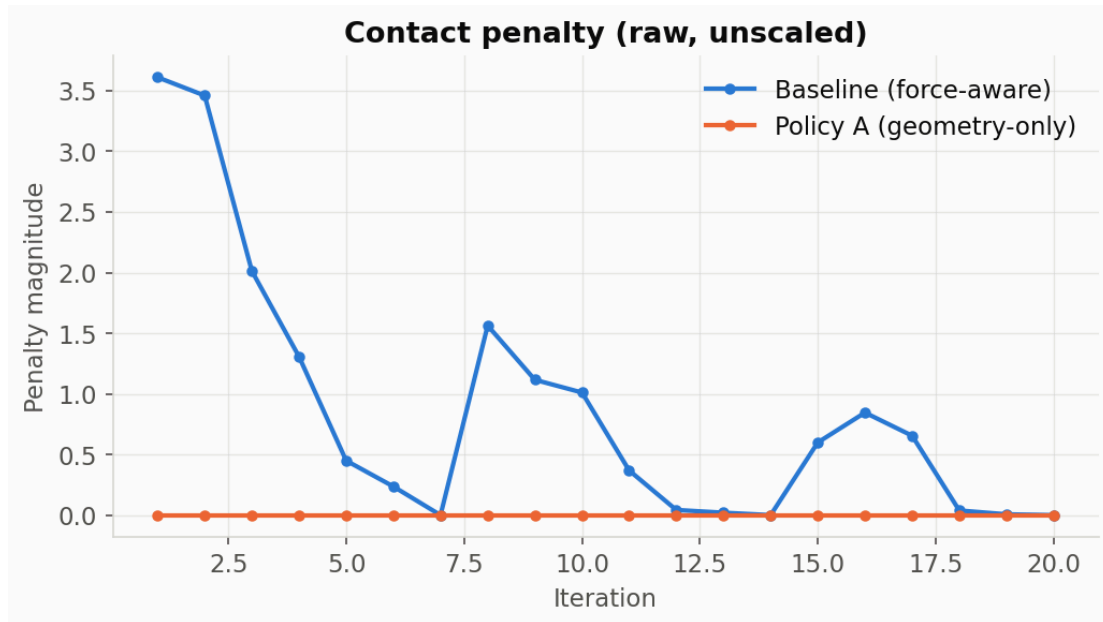
What we see: same qualitative shape as `rewards/iter` for both runs — baseline trending up from strongly negative, Policy A flat and higher. Useful as a sanity check that RL-Games' internal reward pipeline isn't doing something unexpected relative to our env's own reward.

Section 5 is the key result of this report

It is the direct, literal proof that the Policy A ablation removed exactly the signal it was supposed to remove.

5. Contact penalty (raw, unscaled) — the ablation proof

Tag: logs_rew_contact_penalty/iter



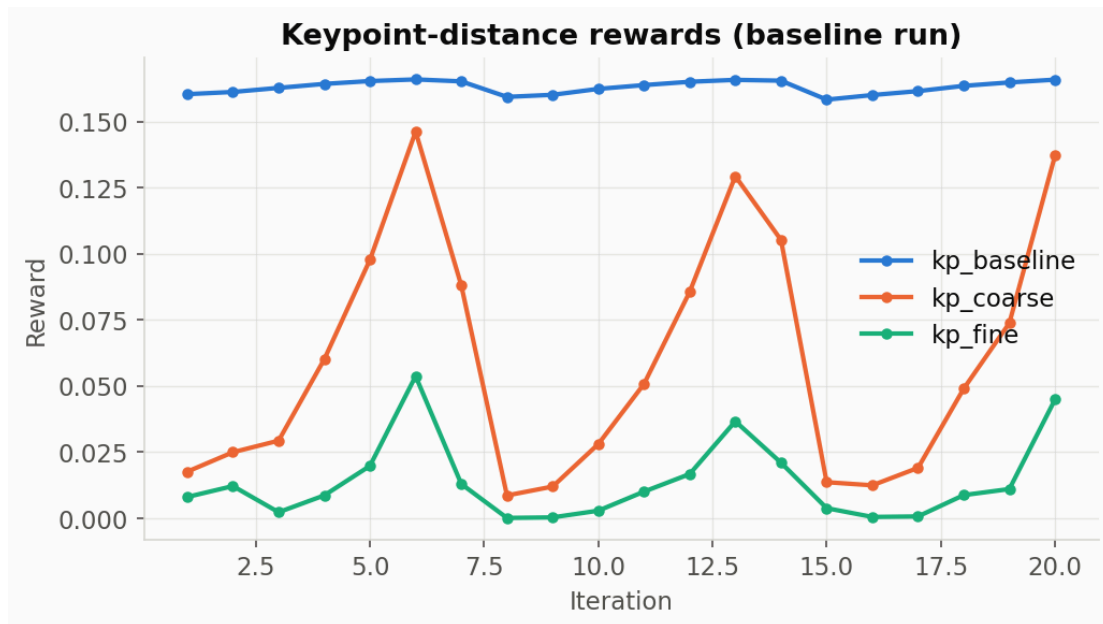
What it is: the raw (unscaled — before multiplying by `contact_penalty_scale`) contact-force penalty term, computed in `ForgeEnv._get_rewards()`:

```
contact_force = torch.linalg.norm(self.force_sensor_smooth[:, 0:3], ord=2, dim=-1)
if self.cfg.use_force_penalty:
    contact_penalty = torch.nn.functional.relu(contact_force -
self.contact_penalty_thresholds)
else:
    contact_penalty = torch.zeros_like(contact_force) # Policy A takes this branch
```

What we see: baseline (blue) fluctuates between roughly 0 and 3.6 as the robot makes and breaks contact during random early exploration. Policy A (orange) sits at exactly 0.0 for all 20 iterations, with zero variance. That flat orange line **is** the deliverable for `next_10_steps.md` step 4 (“confirm the `logs_rew_contact_penalty/force-related` TensorBoard curves are now flat/absent”). An earlier, wrong version of this code would have kept logging the real unscaled contact force even with the reward scale zeroed out — this was specifically fixed so the chart tells the truth.

6. Keypoint-distance rewards (baseline)

Tags: logs_rew_kp_baseline/iter, logs_rew_kp_coarse/iter, logs_rew_kp_fine/iter

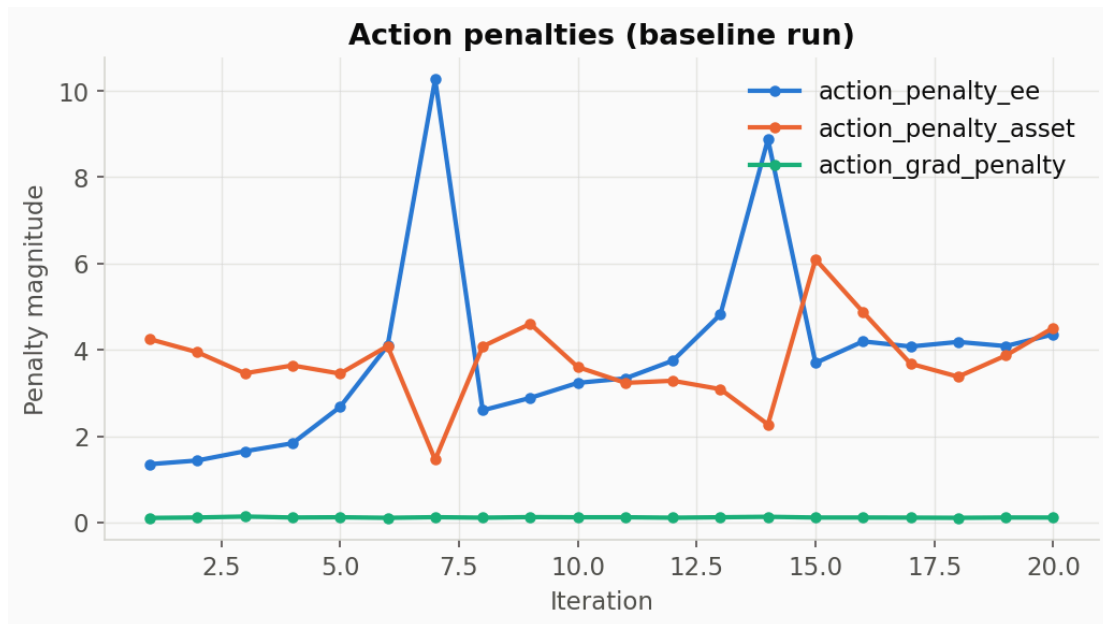


What it is: three reward terms computed from the same underlying quantity — the mean distance between “keypoints” on the held peg and their target positions on the fixed socket (`factory_utils.squashing_fn(keypoint_dist, a, b)` in `factory_env.py`) — each passed through a different squashing curve (`keypoint_coef_baseline`, `_coarse`, `_fine`). The three curves exist so the reward gives useful gradient signal at every distance scale: `kp_baseline` responds across large distances (coarse guidance when the peg is far away), `kp_fine` only lights up once the peg is very close (precision guidance for final insertion). This is upstream `FactoryEnv/ForgeEnv` logic, identical in both our tasks — this chart is baseline-only since Policy A’s geometry observations are unchanged, so this family behaves the same in both runs.

What we see: `kp_baseline` stays nearly flat (~0.16 — it saturates quickly, so it’s not very informative this early). `kp_coarse` and `kp_fine` are noisier and lower in magnitude, spiking on iterations where the random early policy happened to get the peg closer to the target. Expect `kp_fine` in particular to start climbing steadily only once real training has taught the policy to get close enough for it to activate.

7. Action penalties (baseline)

Tags: `logs_rew_action_penalty_ee/iter`, `logs_rew_action_penalty_asset/iter`, `logs_rew_action_grad_penalty/iter`



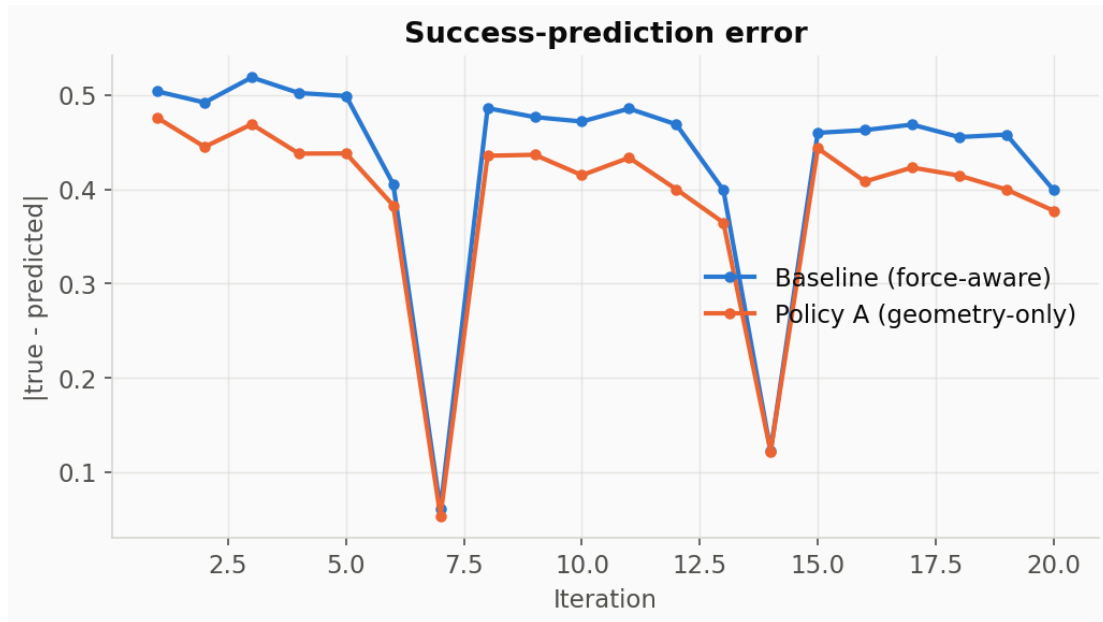
What it is: three different penalties on the policy's raw action output:

- `action_penalty_ee` — L2 norm of the action vector (discourages large commands generally), computed in the `base FactoryEnv`.
- `action_penalty_asset` — a fork-specific (`ForgeEnv`) penalty on position/rotation error relative to the asset-relative action frame.
- `action_grad_penalty` — L2 norm of the **change** between consecutive actions (discourages jerky, high-frequency control), also `base FactoryEnv`.

What we see: `action_penalty_ee` is the largest and most volatile of the three (up to ~10 at iteration 7), since it's an unbounded norm on raw actions from an untrained, near-random early policy. `action_penalty_asset` and `action_grad_penalty` are smaller and noisier. These should generally trend down and smooth out as the policy learns to make smaller, more purposeful, more consistent movements.

8. Success-prediction error

Tag: logs_rew_success_pred_error/iter



What it is: FORGE's distinctive 7th action dimension — the policy outputs a “do I think I've succeeded?” self-prediction (`self.actions[:, 6]`, rescaled from $[-1, 1]$ to $[0, 1]$), and this term penalizes the absolute error between that prediction and the actual ground-truth success flag:

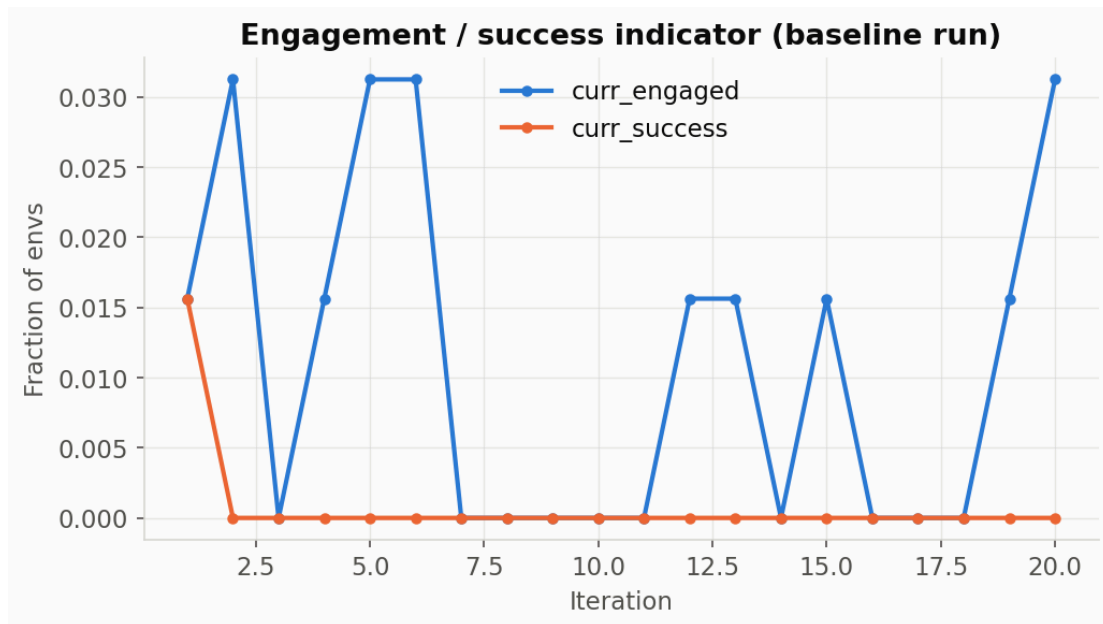
```
success_pred_error = (true_successes.float() - policy_success_pred).abs()
```

Per `ForgeEnv._get_rewards()`, this penalty is only turned on (`self.success_pred_scale = 1.0`) once the true success rate crosses `delay_until_ratio` (0.25) — early in training, with near-zero real successes, this term is computed and logged but not yet weighted into the total reward.

What we see: baseline hovers mostly around 0.4–0.5 (since with a near-random early policy and mostly-zero true successes, a mean prediction near 0.5 gives error near 0.5). Policy A shows the same term (this part of the reward doesn't depend on force observations) with similar behavior. Watch this drop as training progresses and the policy's self-assessment gets calibrated against real outcomes — that's the whole point of early-termination precision/recall (§11).

9. Engagement / success indicator (baseline)

Tags: logs_rew_curr_engaged/iter, logs_rew_curr_success/iter



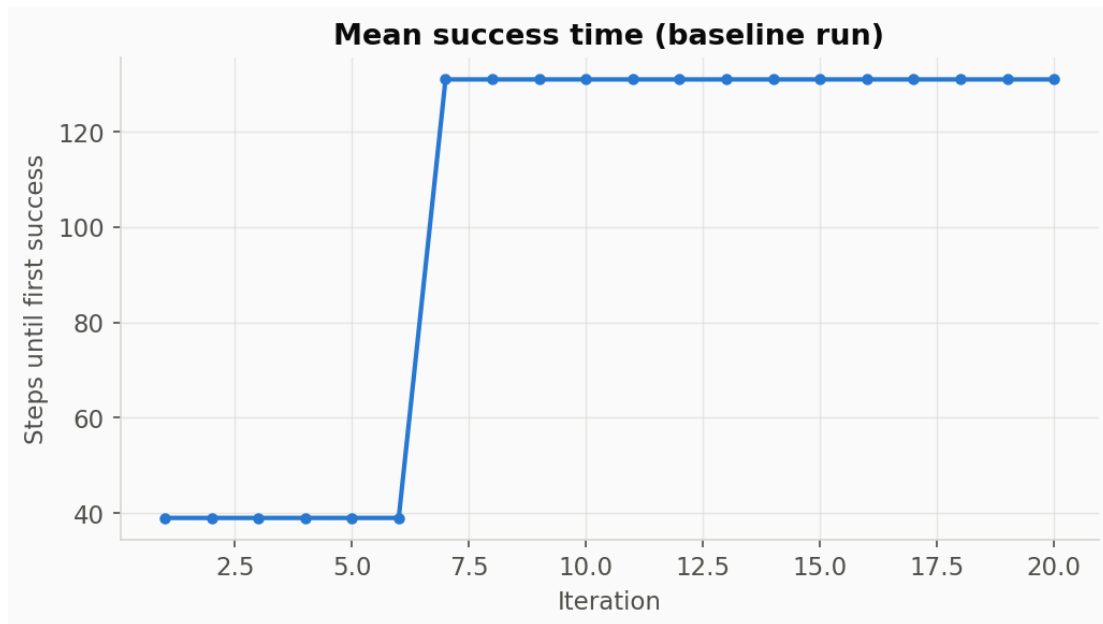
What it is: two boolean-valued (0 or 1, averaged across envs so effectively a fraction) indicators from the base `FactoryEnv`:

- `curr_engaged` — a looser threshold (`engage_threshold`) than full success, meant to catch “peg partially inserted / in contact with socket.”
- `curr_success` — the strict success condition (same one feeding `Episode/Metrics/success_rate` in §1, but logged here as the raw per-step reward term rather than the per-episode aggregate).

What we see: both mostly at 0 with occasional single-env blips ($1/64 = 0.0156$, $2/64 = 0.0312$) — consistent with the near-random early policy occasionally getting lucky. `curr_engaged` should start rising well before `curr_success` does in a real run, since it’s the easier condition to satisfy.

10. Mean success time (baseline)

Tag: `success_times/iter`

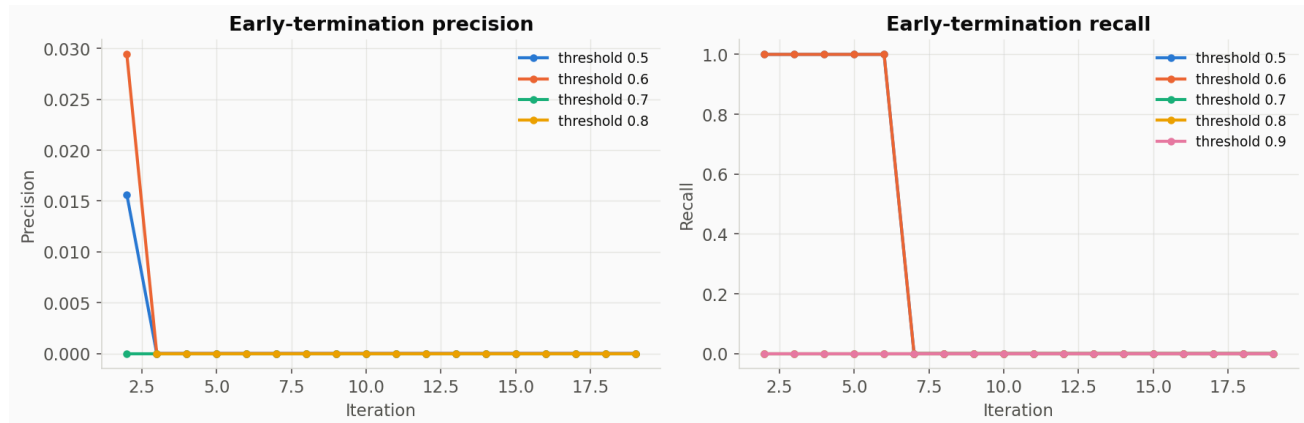


What it is: for episodes that **did** succeed, the mean episode step at which the first success occurred (`self.ep_success_times`, only computed over episodes where `ep_success_times.nonzero()` is non-empty).

What we see: jumps from 39 to 131 partway through the run — but with a 20-iteration smoke test and only 1–2 total successes across all 64 envs, this number is essentially noise from a tiny sample, not a trend. In a real run with many successes per iteration this becomes a genuinely useful “is the policy getting faster at inserting” signal. Not plotted for Policy A here since Policy A had even fewer successes in its smoke test (§1) — the metric wasn’t populated often enough to be worth a comparison at this scale.

11. Early-termination precision & recall (baseline)

Tags: `early_term_precision/{0.5,0.6,0.7,0.8}/iter`, `early_term_recall/{0.5,...,0.9}/iter`



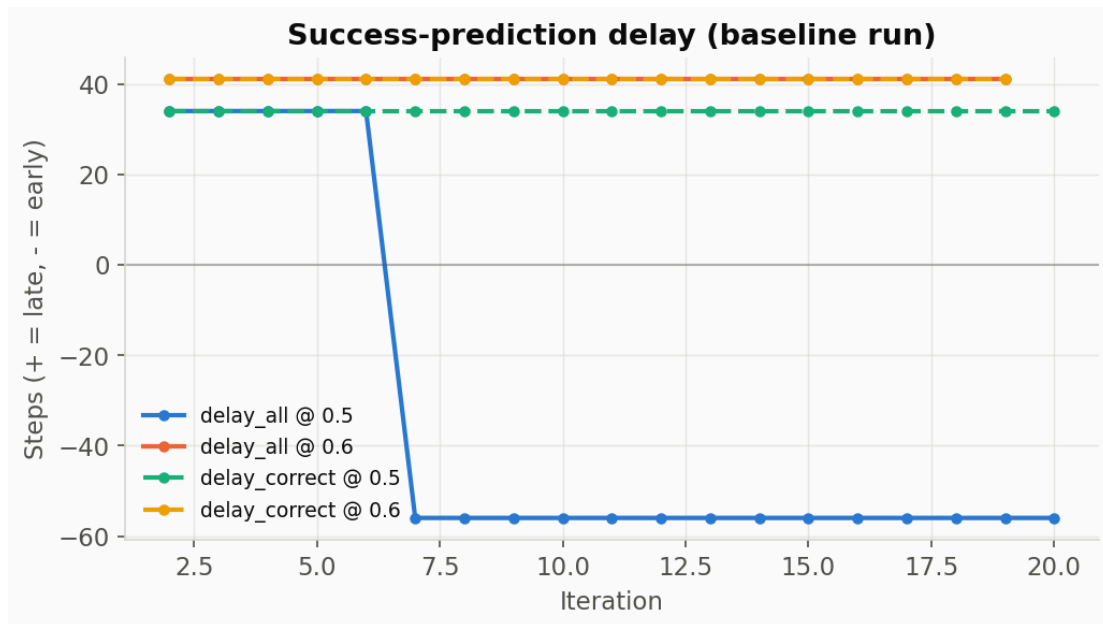
What it is: despite the name, this is **not** about the new `termination_reason` early-termination work — it's an existing FORGE mechanism from `ForgeEnv._log_forge_metrics()` that evaluates the success-prediction head (§8) as a binary classifier at several confidence thresholds. For each threshold t (e.g. 0.5), “predicted success” = the first step where `policy_success_pred > t`. Then:

- **Precision** — of episodes where the policy predicted success at threshold t , what fraction had a real success occur **before** that predicted point?
- **Recall** — of episodes that really did succeed, what fraction did the policy predict (at any point) before or by the real success?

What we see: mostly 0 across both metrics and all thresholds, again because real successes are extremely rare in a 20-iteration smoke test — there isn't enough positive-class data yet for these to be meaningful. This pair becomes useful once training has produced enough real successes to evaluate the self-prediction head's calibration against.

12. Success-prediction delay (baseline)

Tags: `early_term_delay_all/{0.5,0.6}/iter`, `early_term_delay_correct/{0.5,0.6}/iter`

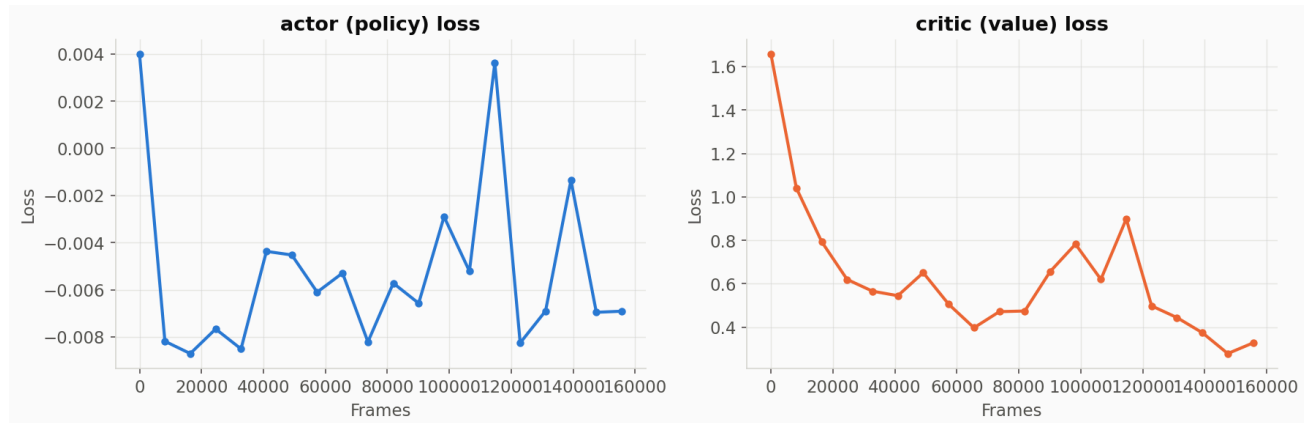


What it is: how many steps **after** (positive) or **before** (negative) the real success moment the policy's self-prediction crossed each threshold — `delay_all` includes every case where both a real and predicted success time exist; `delay_correct` restricts to cases where the prediction came at or after the real success (i.e., excludes cases where the policy “predicted” success too early to be meaningful).

What we see: step changes (e.g. `delay_all/0.5` jumping from 34 to -56 partway through) reflecting which handful of episodes happened to succeed in that iteration — again, noise from a very small sample in a smoke test, not yet a meaningful trend.

13. Actor & critic losses

Tags: losses/a_loss, losses/c_loss



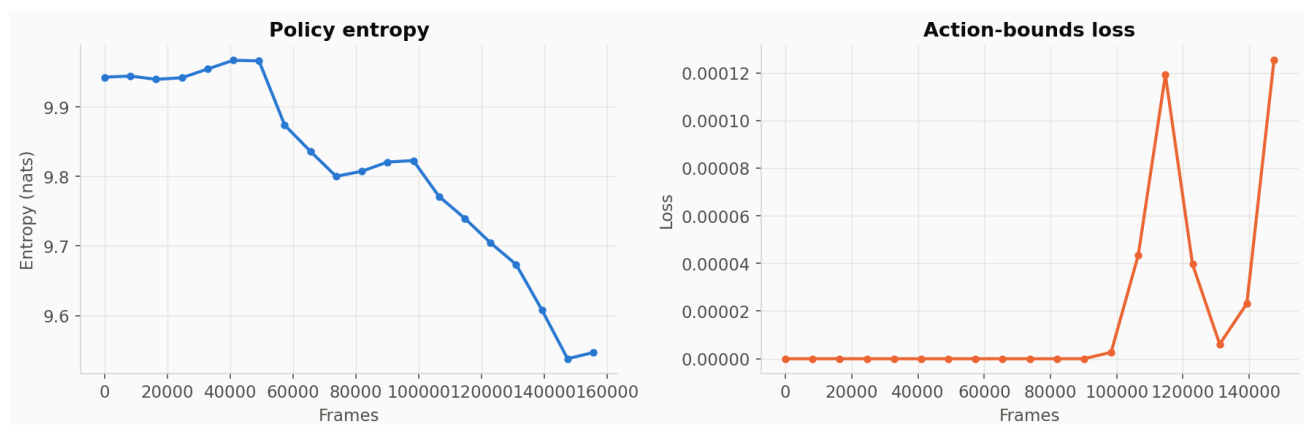
What it is: the two core PPO training losses, logged once per RL-Games training epoch (1 epoch $\approx$ 1 iteration here — 20 epochs total, matching the `--max_iterations 20` smoke test) but plotted against cumulative frame count (0 to 155,648) rather than epoch index, since that's the step value RL-Games attaches to these particular scalars:

- `a_loss` — the clipped-surrogate PPO policy-gradient loss.
- `c_loss` — the value-function (critic) regression loss, predicting expected future return from the 61-dim (baseline) / 57-dim (Policy A) privileged state.

What we see: (baseline) `c_loss` starts high (~ 1.66) and drops sharply to ~ 0.3 – 0.4 within the first $\sim 40,000$ frames — the value function is quickly learning to predict returns better than its random initialization, which is expected and healthy. `a_loss` is much smaller in magnitude and noisier around zero, typical for PPO's clipped objective early in training.

14. Policy entropy & action-bounds loss

Tags: losses/entropy, losses/bounds_loss

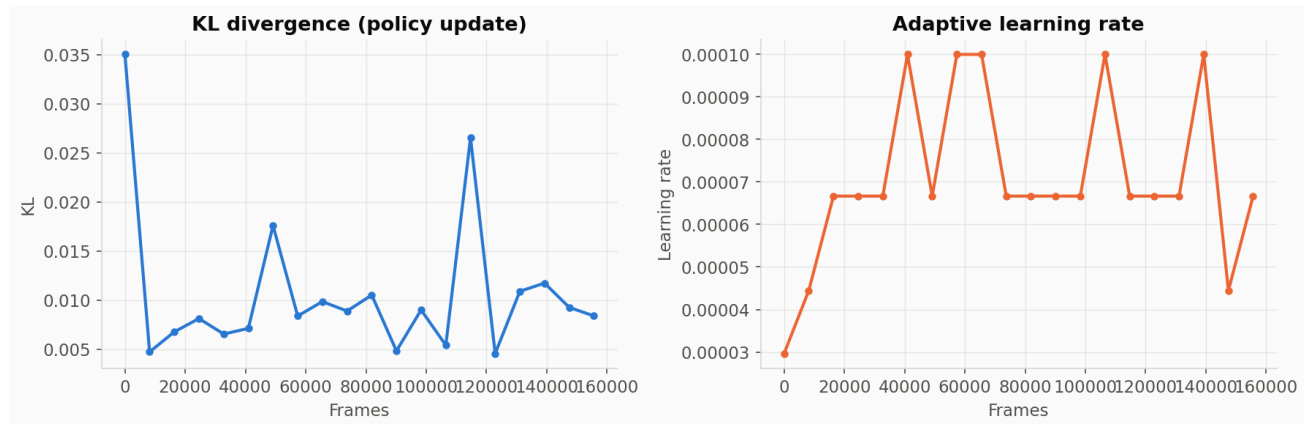


What it is: entropy measures how random/exploratory the policy's action distribution still is (higher = more random); it should trend downward over a full training run as the policy becomes more confident and deterministic, but dropping too fast can mean premature convergence / insufficient exploration. `bounds_loss` penalizes the policy for outputting actions outside the expected $[-1, 1]$ range before squashing.

What we see: entropy starts around 9.95 and drifts down to ~9.5 over the run's ~155,000 frames — a small, expected early dip, nowhere near converged (with 7 continuous action dimensions, max entropy is much higher than the per-dimension case would suggest — 9.5–9.95 nats is a plausible range for a near-random continuous policy this early). `bounds_loss` stays essentially at 0 throughout, meaning the policy isn't yet frequently trying to exceed the valid action range.

15. KL divergence & learning rate

Tags: `info/kl`, `info/last_lr`

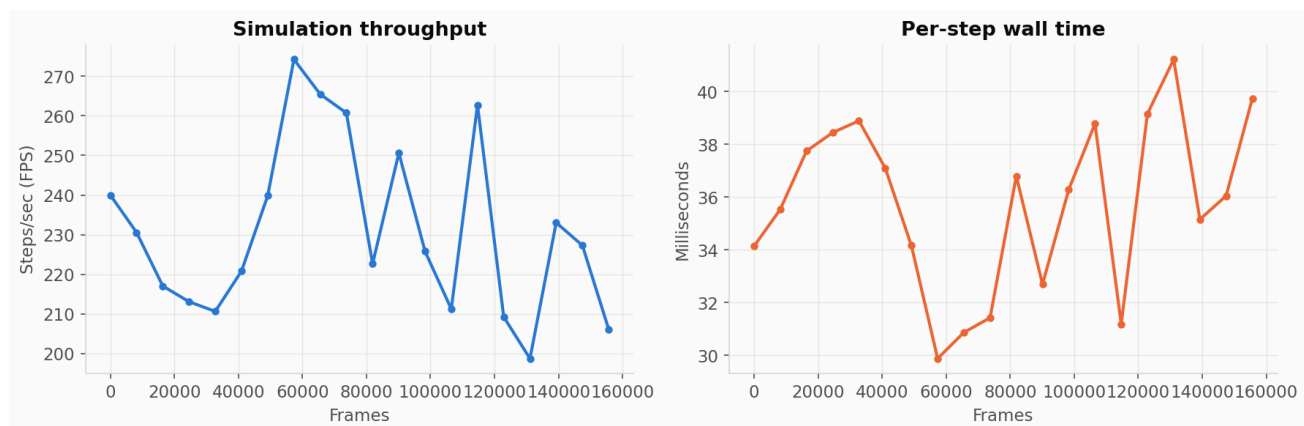


What it is: `kl` is the measured KL-divergence between the policy before and after each PPO update — this is what RL-Games' adaptive learning-rate scheduler actually watches. `last_lr` is that scheduler's current output: when KL creeps above a target band the learning rate is reduced (and vice versa), per the config in `agents/rl_games_ppo_cfg.yaml`.

What we see: `kl` starts with one large spike (~0.035 at frame 0 — normal, first update from network initialization) then settles into a low, noisy band (0.005–0.012). `last_lr` correspondingly jumps around in the $3e-5$ to $1e-4$ range as the scheduler reacts — this back-and-forth is the scheduler doing its job, not a problem.

16. Simulation throughput & step time

Tags: `performance/step_fps`, `performance/step_time`



What it is: pure infrastructure/hardware metrics, unrelated to the policy or reward — `step_fps` is simulated-environment-steps-per-second across all 64 parallel envs; `step_time` is milliseconds per step. Useful for noticing hardware regressions or contention, not for judging training quality.

What we see: roughly 200–270 steps/sec, 34–40 ms per step on this DGX Spark GB10 GPU at `num_envs=64` — expect this to change (not necessarily linearly) at the larger `num_envs=1024` used for real training runs, since GPU parallelism scales differently than a small smoke test predicts.

Tags that exist but aren't graphed here

A few scalar families were skipped as redundant or not yet meaningful at smoke-test scale:

- `info/cval_lr`, `info/e_clip`, `info/epochs`, `info/lr_mul` — mostly-static config values (an unused parallel critic learning rate, the fixed PPO clip range, an epoch counter, and a learning-rate multiplier that stayed at 1.0 throughout both runs) rather than something you'd read a training-health signal off of.
- `performance/rl_update_time`, `performance/step_inference_fps`, `performance/step_inference_time`, `performance/step_inference_rl_update_fps` — finer-grained breakdowns of the same throughput picture as §16, split by which part of the step loop (env step vs. policy inference vs. gradient update) consumed the time. Useful for profiling a real slowdown, not needed for a routine read of training health.

Reproducing these charts

```
# TensorBoard itself (already running on the training machine):
tensorboard --logdir force_peg_rl/logs/rl_games --reload_interval 5 --port 6006
# then open http://localhost:6006/

# Static re-renders of the same data (what generated the PNGs in this report):
python3 docs/assets/tensorboard/gen_charts.py
```